

Research Assignment

Section A

1. A database is a system used to store, organise and manage information so that it can be easily accessed and updated. Databases are important today because businesses and organisations work with large amounts of information and need a reliable way to manage it. For example, banks use databases to store customer accounts and transactions. (IBM, 2026)
2. A data warehouse is a central place where data from different sources is collected and prepared mainly for reporting and analysis. A normal database is usually designed to handle everyday transactions, while a data warehouse is designed for analysing large amounts of historical data. A company would use a data warehouse to combine information from different departments and make better business decisions. (IBM, 2026)
3. A data lake is a large storage system that can keep different types of data, including structured, semi-structured and unstructured data, often in its original form. Unlike a data warehouse, the data does not have to be fully organised before it is stored. A business may prefer a data lake when it has very large amounts of different data and wants to use it later for advanced analytics, machine learning or AI. (IBM, 2024)
4. A data mart is a smaller section of data warehouse that focuses on a specific department or business area. For example, a company could have a sales data mart containing only sales-related information. Data marts are normally used by specific teams such as finance, marketing or sales department because they make it easier for those teams to access information they need. (IBM, 2026)
5. Structured data is data that follows a fixed format and is normally organised into rows and columns. An example would be a customer table containing names, ages, and addresses.

Semi-structured data does not follow a strict table format but still contains some organisation or labels. An example is a JSON file containing customer information.

Unstructured data does not have a fixed structure. Examples include videos, photographs, audio recordings and social media posts. (IBM, 2026)
6. Data modelling is the process of planning how data will be organised and how different pieces of data will relate to each other. It is important because it helps create a database that is organised, accurate and easier to use.
Two types of data models are:
 1. Conceptual data model - gives a high-level view of the main data and relationships.
 2. Logical data model - gives more detail about tables, attributes, and relationships without focusing on the physical storage. (IBM, 2026)
7. Data mining is the process of examining large amounts of data to find useful patterns, relationships and trends. Businesses use it to understand customers and improve decision-making.

For example, a bank can use data mining to identify unusual transactions that could be fraud. A supermarket can analyse customer purchases to discover which products are often bought together. (IBM, 2026)

8. A Database Management System, or DBMS, is a software that allows people and applications to create, manage, search and update databases.

Four examples are:

- MySQL - a relational DBMS commonly used for websites and web applications.
- PostgreSQL - a relational database known for reliability and advanced features.
- Oracle Database - a powerful database commonly used for large organisations.
- MongoDB - a NoSQL database that stores information in flexible documents and is useful when data does not fit neatly into tables.

Each system is chosen depending on the type of data and the needs of the organisation. (IBM, 2026)

9. ETL stands for Extract, Transform and Load.

- Extract means collecting data from different sources
- Transform means cleaning, changing and organising the data into useful format.
- Load means putting the prepared data into a target system such as a data warehouse.

ETL is important because it helps turn raw data from different sources into clean and consistent information that can be used for analysis and reporting. (IBM, 2026)

10. CSV stores data in a simple table-like format separated by commas. It is useful when working with spreadsheets or simple datasets.

- JSON stores information using key-value pairs and is commonly used when exchanging data through APIs.
- Parquet is a column-based format that is useful for large datasets because it is efficient for analytics.
- Avro is a data format that stores data together with its schema and is useful when moving data between systems.

I would use CSV for a small and simple dataset, while Parquet would be more suitable for large analytics datasets. (IBM 2026)

Sources for Section A: IBM and Google Cloud.

Section B

11. Artificial Intelligence, or AI, is technology that allows computers and machines to perform tasks that normally require human intelligence, such as learning, problem-solving and decision-making.

Narrow AI, also called Weak AI, is designed to perform a specific task. An example is a voice assistance such as Siri.

General AI, also called Strong AI, would be able to perform a wide range of intellectual tasks like a human. True General AI does not currently exist. Therefore, an example would be hypothetical rather than an existing system. (IBM, 2026)

12. Machine learning is a part of AI where computers learn patterns from data and use those patterns to make predictions or decisions.

Traditional programming usually requires a programmer to give the computer specific instructions. With machine learning, the computer learns from examples and data instead.

The three main types are:

- Supervised learning - the model learns from labelled data. For example, predicting whether an email is spam.
- Unsupervised learning - the model looks for patterns in data without given labels. For example, grouping customers according to their behaviour.
- Reinforcement learning - the system learns through trial and error by receiving rewards or penalties for its actions. (IBM, 2026)

13. Deep learning is a branch of machine learning that uses neural networks with many layers. It is especially useful for working with complicated data such as images, speech and text.

Neural networks are made up of connected nodes that process information in layers. The input layer receives the information, hidden layers process patterns in the data and the output layer produces the results. (IBM, 2026)

AI → Machine learning → Deep learning

14. Natural Language Processing, or NLP, is a branch of AI that allows computers to understand and work with human language.

Three everyday examples are:

1. Chatbots that understand questions and provide answers.
2. Voice assistance that turn spoken language into commands.
3. Translation applications that translate text from one language into another.

NLP can also be used for sentiment analysis, where a system determines whether text is positive, negative or neutral. (IBM, 2026)

15. A Large Language Model, or LLM, is an AI model trained on a very large amounts of text so that it can understand and generate human-like language.

A standard machine learning model may be designed for one specific task, while an LLM can perform many language-related tasks such as answering questions, summarising information and generating text.

Two examples are:

- GPT - used for tasks such as answering questions, writing and coding.
- Gemini - used for conversation, writing, coding and other AI tasks. (IBM, 2026)

16. Generative AI is AI that can create new content based on patterns it learned from its training data. It can generate things such as text, images, audio, video and computer code.

At a high-level, the model is trained on large amounts of data and learns patterns in the data. When a user provides a prompt, the model uses what it has learned to produce a new response.

Three examples are:

- ChatGPT - generates and works with text.
- Gemini - can generate and work with different types of information
- Midjourney - generates images from text prompts. (IBM, 2026)

17. A training dataset is a collection of information used to teach a machine learning or AI model how to perform a task.

The quality of the training data is important because the model learns from the data. If the data contains errors, missing information or unfair bias, the model can learn those problems and produce unreliable or unbiased results.

For example, if an AI system is trained mostly on information from one group of people, it may not work equally well for other groups.

18. Computer Vision is an area of AI that allows computers to analyse and understand images and videos. A computer does not literally see like a human; instead it processes visual information and identifies patterns, objects or features.

Two real-world examples are:

1. Facial recognition, which can verify or identify a person.
2. Medical imaging, where AI can help identify patterns in scans and assist medical professionals. (IBM, 2026)

Sources for Section B : IBM and Google Cloud.

Section C

19. An API, or Application Programming Interface, allows two different software systems to communicate with each other.

A simple example is a restaurant. The customer tells the waiter what they want, the waiter takes the request to the kitchen and then brings the food back. The waiter is similar to an API because it passes information between the customer and the kitchen.

APIs are important because they allow different applications and services to work together without having to build everything from scratch. (Google Cloud, 2026)

20. An API key is a code used to identify or authenticate an application when it accesses an API.

API keys should be kept private and should not be posted publicly on websites, GitHub or social media. They should preferably be stored using secure environment variables or secret-management tools.

If an API key is exposed, another person may use the account or service and could cause unexpected charges or access information they should not have access to. (Google Cloud, 2026)

21. REST stands for Representational State Transfer. A REST API is a common way for applications to communicate over the internet using HTTP.

The main methods are:

- GET - retrieves information.
- POST - creates or sends new information.
- PUT - updates existing information.
- DELETE - removes information. (Google Cloud, 2026)

22. JSON stands for JavaScript Object Notation. It is a simple format used to store and exchange information.

It is popular with the APIs because it is easy for both humans and computers to read. (Google Cloud, 2026)

23. Prompt engineering is the process of creating clear and specific instructions for an AI tool so that it produces a more useful answer.

Poor prompt:

"Tell me about data."

Improved prompt:

"Explain what data analytics is to a beginner and give three examples of how businesses use it."

Poor prompt:

"Make a graph"

Improved prompt:

"Create a simple bar chart showing monthly sales from January to June and explain which month had the highest sales."

The improved prompts provide more information, which makes it easier for the AI to understand what is required. (IBM, 2026)

24. A MAP function applies the same operation to every item in a list or collection.

For example, in Python:

```
numbers = [1, 2, 3, 4]
```

```
squared = list(map(lambda x: x * x, numbers))
```

```
print(squared)
```

The result would be:

```
[1, 4, 9, 16]
```

The MAP function takes each number and squares it.

25. Cloud computing means using computing resources such as storage, servers and software over the internet instead of having to manage all the physical equipment yourself.

The three main cloud service models are:

- IaaS (Infrastructure as a Service) - provides computing infrastructure such as virtual machines and storage. Example: Amazon EC2.
- PaaS (Platform as a Service) - provides a platform for developers to build and run applications without managing the underlying infrastructure. Example: Google App Engine.
- SaaS (Software as a Service) - provides complete software through the internet. Example: Microsoft 365.

IaaS gives the user more control, while PaaS and SaaS allow the provider to manage more of the underlying system. (AWS, 2021)

26. A Version Control System records changes made to files so that developers can keep track of their work and return to previous versions if necessary.

Git is one of the most widely used version control systems. It is useful to data analysts and developers it allows them to keep track of changes to code, SQL queries, notebooks and other project files.

A commit is a saved record of changes made to a project. A branch is a separate line of development that allows someone to work on changes without immediately affecting the main project. (Git, 2021)

27. A Jupyter Notebook is an interactive environment where users can write and run code, view results and add explanations in the same document.

It is most commonly associated with Python.

Jupyter Notebooks are popular with data analysts and data scientists because they combine code, data analysis, graphs and written explanations in one place.

28. Python is a programming language known for being relatively easy to read and learn. It is very popular in data analytics and

AI because it has many libraries that make data processing, visualisations and machine learning easier.

Four commonly used Python libraries are:

- Pandas - used for working with and analysing tables of data.
- NumPy - used for numerical calculations and arrays.
- Matplotlib - used to create graphs and visualisations.
- Scikit-learn - used for machine learning tasks such as classification, regression and clustering.

Source for Section C: Google Cloud, AWS and Git documentation.

Section D

29. Data privacy is about protecting a person's personal information and making sure it is collected, stored and used appropriately.

It is important because personal information can be misused, stolen or shared without permission.

The GDPR is a data protection law from the European Union. It gives people rights over their personal information and places requirements on organisations that process personal data. The rules can also apply to organisations outside the EU if they offer goods or services to people in the EU or monitor their behaviour there. (European Commission, 2018)

30. Data Ethics is about using data and AI in a responsible and fair way.

Three examples of unethical data use are:

1. Using someone's personal information without permission - this violates their privacy and could cause harm.
2. Using AI to discriminate against certain groups - this is unfair because people could receive different treatment based on biased data.
3. Manipulating data to get the desired result - this can mislead people and lead to poor business or social decisions.

31. Data Governance is a system of rules, processes and responsibilities used to manage data properly within an organisation.

Important parts of good data governance include:

- Data quality - making sure information is accurate and reliable.
- Data security - protecting information from unauthorised access.
- Data privacy - making sure personal information is handled properly.
- Data ownership - deciding who is responsible for particular data.
- Data access - making sure the right person can access the right information.
- Data standards - creating consistent rules for data is used and used. (IBM, 2016)

32. AI bias happens when an AI system produces unfair or unbalanced results because of problems in the data, design

or way the system is used.

For example, if an AI hiring system is trained using historical hiring data that mostly selected men for certain jobs, the system could learn that pattern and unfairly rank male applicants higher.

The consequences could include discrimination, people being denied opportunities and a loss of trust in the AI system. AI systems should therefore be tested for bias and designed with fairness in mind. (IBM, 2026)

33. Role	Main responsibility	Common skills/tools
Data Analyst	Studies data and creates reports and insights for business decisions.	SQL, Excel, PowerBI, Python
Data Scientist	Uses statistics & machine learning to find patterns and make predictions.	Python, SQL, statistics, machine learning
Data Engineer	Builds and maintains systems and pipelines that collect and prepare data.	SQL, Python, databases, cloud platforms
AI Engineer	Builds and implements AI systems and machine learning solutions.	Python, machine learning, APIs, cloud AI tools

34. Business Intelligence, or BI, involves using data, analysis and visualisations to help businesses make better decisions.

Two popular BI tools are PowerBI and Tableau. These tools can connect to data, create reports, and dashboards, and display information through charts and other visualisations. This makes it easier for businesses to identify trends, problems and opportunities. (Microsoft, 2026)

Sources for Section D: European Commission, NIST, IBM and Microsoft

Section E

35. The concept I found most interesting was Artificial Intelligence and especially Generative AI. I found it interesting because AI can now create text, images, and other types of content from a simple instruction. It also showed me how important data is because AI systems learn patterns from large amounts of information.

36. The concept I found most difficult was the difference between a data lake, data warehouse and database. At first they seemed very similar because they all store data. After researching them, I understood that they are designed for different purposes. A database is mainly used for storing and managing operational data, while a warehouse is focused more on organised analysis and a data lake can store large amounts of different types of raw data. (IBM, 2026)

37. I think Data and AI will have a big effect on my future career because many businesses now use data to make decisions. As someone learning data analytics, I can use tools such as Excel, SQL, and PowerBI to analyse information and find useful patterns. AI can also help with tasks such as summarising information, finding patterns and automating repetitive work. In everyday life, I already see AI in search engines, recommendation systems, chatbots and voice assistants.

Sources for Section E: IBM and Microsoft.

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